to running interactive art in Spain and New York, checking runways for debris in Turkey, inspecting labels on products in factories around the world and for rapid face detection in Japan. *[Source: http://opencv.org]*

There is a Yahoo Groups forum in which users can post questions and discussions at *http://groups.yahoo.com/group/OpenCV*; it has about 20,000 members. OpenCV can be executed successfully, without any issue, on Windows, Android, Maemo, FreeBSD, OpenBSD, iOS, Blackberry, Linux and OS X. The official releases of OpenCV can be obtained from SourceForge.

Figure 1: OpenCV

The Statistical Machine Learning library used by OpenCV

- Boosting (meta-algorithm)
- Decision tree learning
- Gradient boosting trees
- Expectation-maximisation algorithm
- k-nearest neighbour algorithm
- Naive Bayes classifier
- Artificial neural networks
- Random forest
- Support vector machine (SVM)

OpenCV is written in C++. The primary interface of OpenCV is in C++. There are full interfaces available in Python, Java and MATLAB/Octave. OpenCV was developed for computational efficiency with a strong focus on real-time applications. One of the goals of OpenCV is to offer an easy-to-use computer vision infrastructure that helps users to build sophisticated vision applications rapidly. The OpenCV library has more than 500 functions that span many areas in computer vision including factory product inspection, medical imaging, security, user interfaces, camera calibration, stereo vision and robotics.

OpenCV—structure and content

OpenCV has broadly five main components, four of which are shown in Figure 2. The CV component contains the basic image processing and higher-level computer vision algorithms; ML is the machine learning library, which includes many statistical classifiers and clustering tools. HighGUI contains I/O routines and functions for storing and loading video and images, and CXCore contains the basic data structures and content.

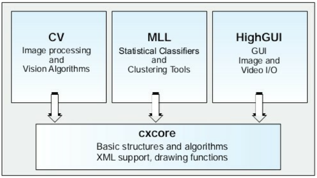

Figure 2: The structure of OpenCV

OpenCV documentation

OpenCV's documentation wiki is more up-to-date than the HTML pages that ship with it, and it also features additional content. The Wiki is located at *http://opencvlibrary.SourceForge.net*. It includes information on:

- Instructions on compiling OpenCV using Eclipse IDE
- Face recognition with OpenCV
- Video surveillance library tutorials
- Camera compatibility
- Links to the Chinese and the Korean user groups
- Stereo correspondence
- View point morphing of cameras
- 3D tracking in stereo
- Eigen object (PCA) functions for object recognition
- Embedded hidden Markov models (HMMs)

Using OpenCV

After installing the OpenCV library, first execute some simple algorithms, for which you have to set up the programming environment. In Visual Studio, create a project and configure the setup so that:

(a) The libraries *highgui.lib*, *cxcore.lib*, *ml.lib*, and *cv.lib* are linked, and

(b) The preprocessor will search the OpenCV *.../opencv/*/include* directories for header files.

These 'include' directories will typically be named as something like *C:/program files/opencv/ cv/include, .../opencv/cxcore/include, .../opencv/ml/include*, and *.../opencv/otherlibs/ highgui.*

Once you have done this, create a new C file for the first program.

To display a digital image loaded from the disk

OpenCV provides the utilities to access a wide array of image file types as well as from videos and cameras. Such utilities are part of a toolkit called HighGUI, which is included in the OpenCV package.

We will make use of some of these utilities to create a